INORGANIC CUMULATIVE EXAM
MARCH 9, 2001

A. Inorganic Structure Engineering

1. Framework materials often use metal ion coordination as components of the framework.

a. Give three examples of metal ions that generally have coordination number two.

b. Give three examples of metal ions that generally have coordination number four (planar).

c. Give three examples of metal ions that generally have coordination number four (tetrahedral).

d. Give three examples of metal ions that generally have coordination number six.

2. Sketch (all atoms and bonds) two examples of multi-dentate ligands that can be used as "connectors" between two metal ions.

3. Many framework systems involve hydrogen bonding.

a. In an X-H…Y H-bond, what atoms are possible for X? What atoms are possible for Y?

b. Describe two ways the presence of H-bonding be determined experimentally. Give the method and explain how H-bonding is shown.

c. Sketch a molecule that can form more than one H-bond simultaneously with another molecule of the same kind.

B. Inorganic Structure Determination

1. Many people only look at the crystallographic R factor to see how well a structure determination turned out.

a. What's the definition of R?

b. What is a typical R factor for a refinement that is considered "quite good"?

2. In a metal complex the metal atom is often the largest atom, yet on the ORTEP illustration of the structure the ellipsoid representing the metal is often the smallest. Explain.

3. How is it possible that X-ray reflection data for which each reflection intensity is only measured with an accuracy of 3-5 percent can give a structure with bond distances that have uncertainties of just a few tenths of one percent?